

TSM-TECHNOVA 2026: INNOVATION SUMMARY

Project Title: CloudPulse: Multi-Cloud Cost Reclamation & Instant Hydration Engine

Team Name: ARGUS Innovators

Problem Statement: Enterprises lose over \$17 Billion annually due to idle, non-production cloud resources. Existing advisory FinOps platforms generate static reports that developers ignore due to outage risks.

Proposed Solution: CloudPulse is an autonomous infrastructure lifecycle engine that monitors idle AWS, GCP, and Kubernetes assets. It safely pauses unutilized workloads and executes ghost resource sweeping (purging unattached storage and orphan static IPs).

Use of Artificial Intelligence: CloudPulse uses predictive multi-signal telemetry analytics (CPU, network bandwidth, DB connections) and time-series evaluation algorithms to detect true idle states without false positives, predicting developer usage windows.

Innovation & Uniqueness: Combines multi-signal AI evaluation, a sub-3-second instant hydration protocol (Web UI & Slack ChatOps), and C-DAC VEGA RISC-V edge hardware co-design with verified carbon footprint reduction accounting.

Target Users & Impact: Enterprise DevOps, FinOps teams, and cloud engineers. Delivers 45% cloud cost reclamation, 100% production uptime, and verifiable carbon emission reductions (kg CO2 avoided).

Current Stage: Fully functional Working Prototype with complete FastAPI backend engine and Next.js 14 interactive dashboard.